Software Project Management & Implementation Plan (SPMP)

Sentra-X: An Evidence-Grounded and Uncertainty-Aware Human-in-the-Loop SOC Alert Triage Copilot

Document Attribute	Details
Document Version	1.1
Release Date	October 12, 2026 (Official Deliverable #3 Deadline)
Academic Term	Fall Semester 2026 (Teaching: 17 August 2026 – 27 November 2026; Final Exams: 30 November 2026)
Institution	Department of Computer and Information Systems Engineering, NED University of Engineering & Technology
Authors / Project Team	Asad Ullah Zafar (CS-24082), Atta Ur Rehman (CS-24092), Abdur Rehman (CS-24098)
Submitted To / Supervisors	Mr. Muhammad Ali Akhtar — Practical Class Teacher (DDL Lab, Mr. Khurram Alvi) & Dr. Fakhra Aftab — Theory Class Teacher
Companion Specification	Sentra-X Software Requirements Specification (SRS v1.3)

Table of Contents

1. Project Overview
    - 1.1 Purpose of the Project Plan
    - 1.2 Project Scope & Core Novelty Pillars
    - 1.3 Strict Out-of-Scope Boundaries
    - 1.4 Official Fall Semester 2026 Deliverables & Deadlines
2. Process Model & Engineering Methodology
    - 2.1 Selected SDLC Model: Iterative & Incremental Agile-Scrum
    - 2.2 Rationale for Methodology Selection
    - 2.3 Sprint & Supervisor Review Cadence
3. Project Organization & Team Responsibilities
    - 3.1 Team Structure & Module Ownership
    - 3.2 Module Distribution Matrix (Core Features & Subsystem Split)
    - 3.3 Responsibility Assignment Matrix (RACI)
4. Work Breakdown Structure (WBS) & Technical Work Packages
    - 4.1 WBS Architecture Diagram
    - 4.2 Detailed Work Packages (`WP-1` to `WP-6`)
    - 4.3 Repository & Module Implementation Blueprint
5. Project Schedule, Sprints & Milestones (15-Week Fall Semester)

    - 5.1 Fall Semester 2026 Master Schedule (Weeks 1–15: Aug 17 – Nov 27, 2026)
    - 5.2 Critical Path & Parallel Execution Diagram
    - 5.3 Milestone Gate Criteria (`M1` to `M6`)
6. Resource, Effort & Cost Estimation
    - 6.1 Hardware & Software Resource Requirements
    - 6.2 Engineering Effort Estimation (Person-Hours Across 15 Weeks)
    - 6.3 LLM API Quota & Operational Budget Control
7. Risk Management Plan
    - 7.1 Risk Assessment Methodology
    - 7.2 Technical, Operational & Schedule Risk Register
8. Quality Assurance, Verification & Evaluation Plan
    - 8.1 Multi-Layer Testing Strategy (For Nov 25 Testing Document)
    - 8.2 Non-Functional Target Verification Matrix
    - 8.3 Research & Adversarial Evaluation Protocol (`FR-10`)
9. Configuration, Change & Delivery Management
    - 9.1 Version Control & Branching Strategy
    - 9.2 Documentation & Persistent Context Protocol
    - 9.3 Final CD Submission & Module Distribution Handover Checklist (Nov 25–27)
10. Appendix: Document Revision History

---

1. Project Overview

1.1 Purpose of the Project Plan

This Software Project Management & Implementation Plan (SPMP) defines the execution roadmap, work breakdown structure (WBS), team member module distribution, 15-week Fall Semester schedule, resource estimates, risk mitigations, and verification protocols for **Sentra-X**. It translates the functional (`FR-1`–`FR-10`), non-functional (`NFR-1`–`NFR-6`), and relational data requirements specified in the **Sentra-X Software Requirements Specification (SRS v1.3)** into an actionable engineering plan aligned with the official **Fall Semester 2026 Project Deliverables & Deadlines (17 August 2026 – 27 November 2026)**.

1.2 Project Scope & Core Novelty Pillars

Security Operations Centers (SOCs) face severe alert fatigue from high-volume SIEM platforms such as **Wazuh**, while naive LLM triage tools suffer from hallucinated threat attributions, overconfident binary verdicts on incomplete telemetry, silent downgrades of critical alerts, and vulnerability to indirect prompt injection.

Sentra-X is a containerized, human-in-the-loop decision-support copilot built with **React 19 (TypeScript + Vite + Tailwind CSS + shadcn/ui)**, **PostgreSQL 16**, **Python 3.11+ (FastAPI + SQLAlchemy 2.0)**, and a swappable **Google Gemini / Anthropic Claude** LLM adapter. Every engineering milestone reinforces Sentra-X's **five core novelty pillars**:

1. **Pillar 1 — Explicit Evidence Grounding (`FR-2`):** Compiles a structured `EvidencePack` combining raw Wazuh telemetry, MITRE ATT&CK metadata, host criticality tier (`FR-9`), and up to `K = 5` temporally correlated alerts (`24h` window). Requires the LLM to return **dot-notation JSON key paths** (`cited_evidence_keys`) that are deterministically verified against the `EvidencePack`.

2. **Pillar 2 — Uncertainty-Aware Abstention (`FR-3`):** Refuses to force binary `TRUE_POSITIVE` / `FALSE_POSITIVE` classifications when evidence is weak (`confidence_score < 0.60`), citations are empty/invalid, outputs are self-contradictory, or prompt injection is suspected alongside a benign recommendation—deterministically routing the alert to `NEEDS_ANALYST_REVIEW`.
3. **Pillar 3 — Deterministic Multi-Tier Safety Floor (`FR-3`):** Enforces a hard backend policy blocking AI false-positive dismissals or severity downgrades on Critical (`rule.level ≥ 12`) and High (`rule.level 10–11`) Wazuh detections.
4. **Pillar 4 — Dynamic Few-Shot Feedback Calibration (`FR-4`):** Stores verified human override rationales (`≥ 10 characters`) in PostgreSQL and dynamically injects the top `2` matching active exemplars in `< 50ms` using two-tier signature matching (`rule_id + process_signature`, then `rule_id + asset_criticality`), with Team Lead exemplar revocation (`is_active = false`).
5. **Pillar 5 — Adversarial Prompt-Injection Defense (`FR-2`, `NFR-3`):** Sanitizes control characters and delimiter spoofing tokens (`PromptSanitizer`), flags `injection_suspect`, and isolates untrusted telemetry inside read-only JSON envelopes to achieve a **`0%` bypass rate** across a 30-payload adversarial benchmark.

1.3 Strict Out-of-Scope Boundaries

To guarantee full working completion before the **25th November 2026** final submission deadline, the following items are strictly out of scope:

- **No Autonomous Active Response:** Sentra-X is a read-only decision-support system; it never modifies Wazuh rules, kills processes, blocks IPs, or alters firewall policies.
- **No LLM Weight Fine-Tuning or GNNs:** Adaptation relies strictly on in-context few-shot exemplar retrieval (`FR-4`); parameter fine-tuning (LoRA/SFT), multi-agent architectures, and Graph Neural Networks are excluded.
- **No Orphaned Alert States:** Every alert transitions strictly within the closed-loop 5-state machine (`PENDING_REVIEW`, `AWAITING_ANALYST`, `NEEDS_MANUAL_TRIAGE`, `ESCALATED`, `CLOSED`).

1.4 Official Fall Semester 2026 Deliverables & Deadlines

Per the official course schedule (**Teaching: 17 August 2026 – 27 November 2026 | Final Examinations Begin: 30 November 2026**), all project deliverables are submitted to **Mr. Muhammad Ali Akhtar (DDL Lab, Mr. Khurram Alvi)**:

S.No	Official Deliverable	Official Deadline	Submitted To	Current Status
1	**Project Proposal**	**07th September, 2026 (Monday)**	Mr. Muhammad Ali Akhtar (DDL Lab Mr. Khurram Alvi)	**Completed** (`Sentra-X_Proposal_Updated.pdf`)
2	**Software Requirement Specification (SRS)**	**28th September, 2026 (Monday)**	Mr. Muhammad Ali Akhtar (DDL Lab Mr. Khurram Alvi)	**Completed** (`sentraX_srs.md` / `SentraX_SRS_Final.pdf`)

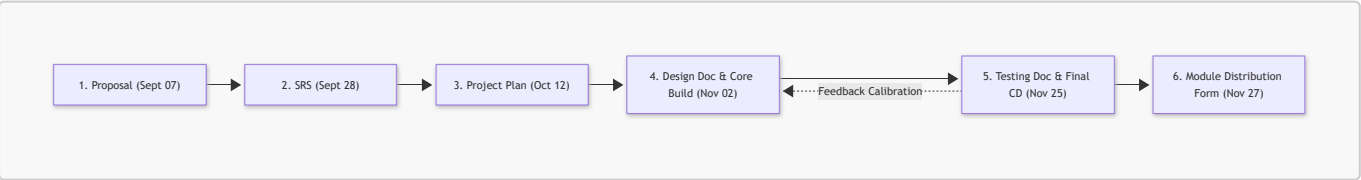
S.No	Official Deliverable	Official Deadline	Submitted To	Current Status
3	**Project Plan**	**12th October, 2026 (Monday)**	Mr. Muhammad Ali Akhtar (DDL Lab Mr. Khurram Alvi)	**Completed** (`sentraX_project_plan.md`)
4	**Design Document**	**02nd November, 2026 (Monday)**	Mr. Muhammad Ali Akhtar (DDL Lab Mr. Khurram Alvi)	In Progress / Scheduled (Weeks 7–11)
5	**Testing Document & Final Project (in CD)**	**25th November, 2026 (Wednesday)**	Mr. Muhammad Ali Akhtar (DDL Lab Mr. Khurram Alvi)	Scheduled (Weeks 12–15)
6	**Module Distribution Form, Core Features, Working & Non-Working Functionalities Details**	**27th November, 2026 (Friday)**	Mr. Muhammad Ali Akhtar (DDL Lab Mr. Khurram Alvi)	Scheduled (Week 15 Final Handover)

---

2. Process Model & Engineering Methodology

2.1 Selected SDLC Model: Iterative & Incremental Agile-Scrum

Sentra-X follows an **Iterative and Incremental Agile-Scrum** lifecycle aligned directly with the 15-week Fall Semester teaching window (`17 August 2026 – 27 November 2026`). Following the completion of the Proposal (`Sept 07`), SRS (`Sept 28`), and Project Plan (`Oct 12`), the remaining 6.5 weeks are structured into three focused engineering sprints mapped to the **Design Document (`Nov 02`)**, **Testing Document & Final Project CD (`Nov 25`)**, and **Module Distribution & Functionality Audit (`Nov 27`)**.



2.2 Rationale for Methodology Selection

1. **Database-First Contract Alignment:** By having **Asad Ullah Zafar** establish the PostgreSQL schema, SQLAlchemy ORM layer, and frontend-backend API integration contracts early, the React dashboard and backend Wazuh/LLM pipelines can be built concurrently without integration bottlenecks.
2. **Parallel Wazuh & Backend Execution:** While Asad builds the Analyst Queue, Evidence Trace Modal, Decision Capture UI, and DB integration, **Atta Ur Rehman** and **Abdur Rehman** develop the Wazuh SIEM ingestion service, Evidence Pack compiler, Prompt Sanitizer, LLM adapters, and Deterministic Safety Floor in parallel.

3. **Deadline Alignment:** Each sprint produces both working software and the exact formal artifact required by **Mr. Muhammad Ali Akhtar** on `Nov 02`, `Nov 25`, and `Nov 27`.

2.3 Sprint & Supervisor Review Cadence

- **Bi-Weekly Practical Lab Checkpoints:** Conducted in **DDL Lab** with **Mr. Muhammad Ali Akhtar** and **Mr. Khurram Alvi**, paired with theory alignment reviews under **Dr. Fakhra Aftab**.
- **Definition of Done (DoD):** A module is marked complete only when: (a) database persistence and API integration are verified end-to-end, (b) RBAC (`FR-8`) and deterministic safety rules (`FR-3`) pass unit/integration tests, and (c) [gemini.md](file:///c:/projects/SentraX/docs/gemini.md) is updated.

---

3. Project Organization & Team Responsibilities

3.1 Team Structure & Module Ownership

Work is distributed across the three team members according to subsystem specialization—separating **Frontend UX + Full-Stack Database Integration** from **Wazuh SIEM Telemetry + Backend AI/Guardrail Services**:

Team Member	Roll No.	Primary Role	Core Engineering Ownership
Asad Ullah Zafar	`CS-24082`	**Frontend, Database & Full-Stack Integration Lead**	**Frontend SPA & UI/UX:** React 19 + TypeScript + Tailwind/shadcn dashboard, Penpot wireframes, Prioritized Analyst Review Queue & Split-Pane Evidence Trace Modal (`FR-5`), Decision Capture UI (`Approve`, `Override`, `Escalate`, `Retry AI`, `Resolve` — `FR-6`), Team Lead Agreement Analytics & CSV Export UI (`FR-7`), Login & Role Guards (`FR-8`), and Asset/Exemplar Governance UI (`FR-4`, `FR-9`). **Database & Integration:** PostgreSQL 16 schema (all 7 tables), SQLAlchemy 2.0 ORM models, composite/partial indexes (`< 50ms` lookup), `analyst_decisions_audit` immutability trigger, and end-to-end Frontend ↔ FastAPI ↔ Database integration.
Atta Ur Rehman	`CS-24092`	**Wazuh SIEM & Core Backend Infrastructure Lead**	**Wazuh Integration & Ingestion (`FR-1`, `FR-9`):** Wazuh v4.x environment setup & alert replay generator, `IngestionService` (REST poller every `30s` + webhook listener), informational noise filter (`rule.level < 4`), `wazuh_alert_id` deduplication, `process_signature` extraction, and Backend Host/Asset Criticality service (`FR-9`). **Backend Security & Calibration Services (`FR-4`, `FR-8`):** Salted `bcrypt` (`cost ≥ 10`) & JWT RBAC middleware (`FR-8`), Backend `FeedbackCalibrationService` (two-tier exemplar retrieval & revocation logic — `FR-4`), and Docker Compose orchestration.

Team Member	Roll No.	Primary Role	Core Engineering Ownership
Abdur Rehman	CS-24098	**Backend AI Triage, Guardrails & Evaluation Lead**	**Wazuh Telemetry Enrichment & Sanitization (FR-2, NFR-3):** EvidenceAssembler (Wazuh telemetry parsing, MITRE ATT&CK mapping, 24h host/IP temporal correlation, and dot-notation key extraction) and PromptSanitizer (control/delimiter stripping, injection_suspect detection, read-only JSON envelope). **LLM Adapters, Safety Floor & Benchmarking (FR-2, FR-3, FR-7, FR-10):** Swappable GeminiAdapter / ClaudeAdapter (10s timeout + 3x retry backoff), Deterministic RulesEngine (Abstention & Multi-Tier Safety Floor — FR-3), Backend Analytics SQL aggregation (FR-7), and 30-Payload Adversarial & Baseline Evaluation Runner (FR-10).

3.2 Module Distribution Matrix (Core Features & Subsystem Split)

This matrix serves as the baseline specification for **Deliverable #6 (Module Distribution Form, due 27th November 2026)**:

Subsystem / Module	Mapped FRs / NFRs	Primary Owner	Secondary / Integration Partner	Target Completion
1. PostgreSQL 16 Schema, ORM Models, Indexes & Audit Immutability Trigger	Section 7, NFR-4	**Asad Ullah Zafar (CS-24082)**	Atta Ur Rehman (CS-24092)	Oct 24, 2026
2. Frontend ↔ Backend ↔ Database REST Contract & State Integration	FR-5, FR-6, FR-7, FR-8, FR-9	**Asad Ullah Zafar (CS-24082)**	Atta Ur Rehman & Abdur Rehman	Nov 18, 2026
3. Prioritized Analyst Review Queue & Split-Pane Evidence Trace Modal (React SPA)	FR-5, Usability NFR	**Asad Ullah Zafar (CS-24082)**	Abdur Rehman (CS-24098)	Nov 14, 2026
4. Analyst & Team Lead Decision Capture UI, Login Screen & Analytics Dashboard	FR-6, FR-7, FR-8 UI	**Asad Ullah Zafar (CS-24082)**	Atta Ur Rehman (CS-24092)	Nov 18, 2026
5. Wazuh SIEM Setup, Poller/Webhook Ingestion, Noise Filter & Deduplication	FR-1, NFR-2, NFR-5	**Atta Ur Rehman (CS-24092)**	Abdur Rehman (CS-24098)	Oct 28, 2026
6. Backend JWT Auth / RBAC Middleware, Asset Criticality Service & Docker Compose	FR-8, FR-9, NFR-3, NFR-6	**Atta Ur Rehman (CS-24092)**	Asad Ullah Zafar (CS-24082)	Nov 02, 2026

Subsystem / Module	Mapped FRs / NFRs	Primary Owner	Secondary / Integration Partner	Target Completion
7. Backend Dynamic Few-Shot Feedback Calibration Service (Tiered Retrieval)	FR-4, NFR-4	**Atta Ur Rehman (CS-24092)**	Asad Ullah Zafar (CS-24082)	Nov 10, 2026
8. Evidence Pack Compiler, Prompt-Injection Sanitizer & Gemini/Claude Adapters	FR-2, NFR-1, NFR-3, NFR-6	**Abdur Rehman (CS-24098)**	Atta Ur Rehman (CS-24092)	Nov 06, 2026
9. Deterministic Abstention & Multi-Tier Safety Floor Rules Engine	FR-3, NFR-1	**Abdur Rehman (CS-24098)**	Atta Ur Rehman (CS-24092)	Nov 10, 2026
10. Backend Analytics Aggregation Queries & 30-Payload Evaluation Runner	FR-7 API, FR-10, NFR-3	**Abdur Rehman (CS-24098)**	Asad Ullah Zafar & Atta Ur Rehman	Nov 22, 2026

3.3 Responsibility Assignment Matrix (RACI)

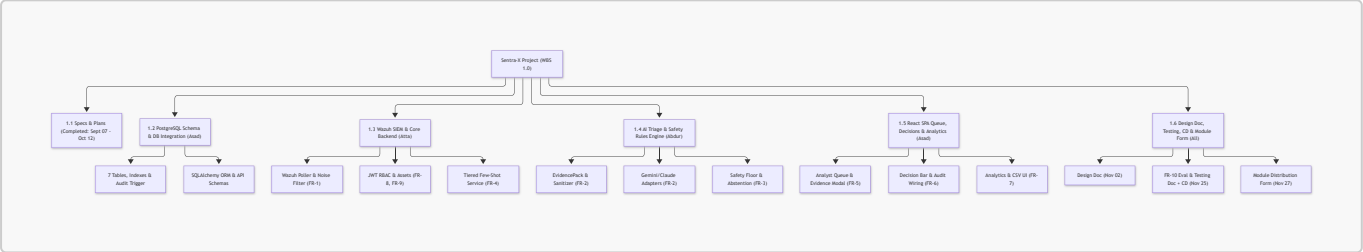
*Legend: **R** = Responsible (Implements), **A** = Accountable (Approves/Owns Quality), **C** = Consulted, **I** = Informed.*

Work Package / Official Deliverable	Asad (CS-24082) *(Frontend & DB Integration)*	Atta (CS-24092) *(Wazuh & Backend Core)*	Abdur (CS-24098) *(Backend AI & Guardrails)*	Supervisors
WP-1: Proposal, SRS & Project Plan (S.No 1–3)	**R / A**	**R**	**R**	**A / I**
WP-2: PostgreSQL Schema, ORM & DB Integration Layer	**R / A**	**C**	**C**	**I**
WP-3: Wazuh Ingestion (FR-1), Auth/RBAC (FR-8), Assets (FR-9) & Calibration (FR-4)	**C** *(DB/API Wiring)*	**R / A**	**R**	**I**
WP-4: Evidence Pack, Sanitizer (FR-2), LLM Adapters & Safety Floor (FR-3)	**I**	**R**	**R / A**	**I**
WP-5: React SPA Queue (FR-5), Decisions (FR-6), Analytics (FR-7) & End-to-End Integration	**R / A**	**C**	**C**	**I**

Work Package / Official Deliverable	Asad (CS-24082) *(Frontend & DB Integration)*	Atta (CS-24092) *(Wazuh & Backend Core)*	Abdur (CS-24098) *(Backend AI & Guardrails)*	Supervisors
WP-6: Design Doc (S.No 4), Testing Doc & CD (S.No 5), Benchmark (FR-10) & Module Form (S.No 6)	**R / A**	**R**	**R**	**A**

4. Work Breakdown Structure (WBS) & Technical Work Packages

4.1 WBS Architecture Diagram



4.2 Detailed Work Packages (WP-1 to WP-6)

WP-1: Proposal, SRS & Project Plan *(Completed: Weeks 1–8 | Aug 17 – Oct 12, 2026)*

- **Owners:** Asad Ullah Zafar, Atta Ur Rehman, Abdur Rehman
- **Completed Tasks:**
  - 1.1.1 Literature review (sentra_novelty_research.pdf) & Deliverable #1 (**Project Proposal**, submitted 07 Sept 2026).
  - 1.1.2 Deliverable #2 (**Software Requirement Specification**, submitted 28 Sept 2026, refined to v1.3 with 10 sections, System Models, ER Schema, and RTM).
  - 1.1.3 Frontend React 19 + TypeScript + Vite + Tailwind CSS + shadcn/ui project initialization (frontend/).
  - 1.1.4 Deliverable #3 (**Project Plan / SPMP**, completed 12 Oct 2026).

WP-2: Database Architecture, ORM Layer & Full-Stack Data Contracts *(Weeks 8–10 | Oct 13 – Oct 26, 2026)*

- **Lead Owner: Asad Ullah Zafar (CS-24082)**
- **Mapped Requirements:** SRS Section 7, FR-4 Index, FR-5 Index, FR-6 Audit Trigger, NFR-4
- **Tasks:**
  - 1.2.1 **PostgreSQL 16 Schema & Migrations:** Implement the 7 core tables (users, assets, alerts, evidence_packs, ai_assessments, analyst_decisions_audit, feedback_exemplars) via SQLAlchemy 2.0 and Alembic.
  - 1.2.2 **High-Speed Indexing & Audit Immutability:** Create the unique index on alerts(wazuh_alert_id), 24h correlation index on alerts(agent_id, src_ip, alert_timestamp DESC), queue index on alerts(status, rule_level DESC, ingested_at

ASC), < 50ms partial composite index on feedback_exemplars, and the PostgreSQL BEFORE UPDATE OR DELETE trigger on analyst_decisions_audit.

- 1.2.3 **Frontend-Backend-Database API Contracts:** Define shared Pydantic v2 response models and TypeScript interfaces (AlertQueueItem, EvidencePackDTO, AIAssessmentDTO, DecisionRequestDTO, AnalyticsSummaryDTO) so the React frontend and FastAPI routers integrate seamlessly with PostgreSQL.

WP-3: Wazuh SIEM Integration, Core Backend Infrastructure & Feedback Calibration *(Weeks 9–12 | Oct 19 – Nov 10, 2026)*

- **Lead Owner: Atta Ur Rehman (CS-24092)** *(with Abdur Rehman on Wazuh log parsing & Asad on DB wiring)*
- **Mapped Requirements:** FR-1, FR-4, FR-8, FR-9, NFR-2, NFR-3, NFR-4, NFR-5
- **Tasks:**
  - 1.3.1 **Wazuh v4.x Connector & Alert Replay Generator (FR-1):** Configure Wazuh REST API polling (every N = 30s) and /api/v1/ingest/webhook endpoint, plus a synthetic Wazuh alert replay script for high-throughput (5 alerts/sec) and offline testing.
  - 1.3.2 **Noise Filtering & Deduplication (FR-1):** Drop informational events (rule.level < 4), deduplicate by wazuh_alert_id, extract normalized process_signature, and persist PENDING_REVIEW alerts.
  - 1.3.3 **JWT Authentication & RBAC Middleware (FR-8):** Implement salted bcrypt (cost ≥ 10) password verification, JWT generation, and FastAPI role guards (Analyst, TeamLead, Admin) returning HTTP 401 / 403.
  - 1.3.4 **Host Criticality Service (FR-9) & Tiered Few-Shot Calibration (FR-4):** Auto-register new Wazuh agent.id entries as MEDIUM criticality, implement < 50ms two-tier exemplar lookup (Tier 1: rule_id + process_signature; Tier 2: rule_id + asset_criticality), and expose Team Lead exemplar revocation endpoints.

WP-4: Guarded AI Triage Pipeline, Prompt Sanitizer & Deterministic Safety Floor *(Weeks 9–12 | Oct 19 – Nov 10, 2026)*

- **Lead Owner: Abdur Rehman (CS-24098)** *(with Atta Ur Rehman on backend worker orchestration)*
- **Mapped Requirements:** FR-2, FR-3, FR-7 (Backend SQL), NFR-1, NFR-3, NFR-6
- **Tasks:**
  - 1.4.1 **Evidence Pack Compiler (FR-2):** Build EvidenceAssembler to combine raw Wazuh alert fields, MITRE ATT&CK IDs/tactics, host criticality_tier, and up to K = 5 correlated alerts (24h window) into a structured EvidencePack with precomputed valid_dot_keys.
  - 1.4.2 **Prompt-Injection Sanitizer (FR-2, NFR-3):** Build PromptSanitizer to strip control bytes and delimiter-spoofing strings (</system>, [INST], code fences), flag adversarial override strings with injection_suspect = true, and isolate telemetry inside a read-only JSON envelope.
  - 1.4.3 **Swappable Gemini & Claude Adapters (FR-2, NFR-6):** Implement GeminiAdapter (google-genai) and ClaudeAdapter (anthropic) behind an environment-configured factory with strict JSON schema validation, 10s timeout, and 3x exponential backoff (1s, 2s, 4s) transitioning failed alerts to NEEDS_MANUAL_TRIAGE.
  - 1.4.4 **Deterministic Abstention & Multi-Tier Safety Floor Engine (FR-3):**

    - Force `effective_verdict = NEEDS_ANALYST_REVIEW` when `confidence_score < 0.60`, `cited_evidence_keys` is empty/unverified against `valid_dot_keys`, verdict/severity is contradictory, or `injection_suspect = true` with a benign recommendation.
    - Enforce Critical Safety Floor (`rule_level ≥ 12` blocks `FALSE_POSITIVE` or `< HIGH` severity) and High Safety Floor (`rule_level 10–11` blocks `FALSE_POSITIVE` or `LOW` severity).

`WP-5`: Frontend React SPA — Analyst Queue, Evidence Modal, Decision Capture & Analytics Integration *(Weeks 10–14 | Oct 26 – Nov 18, 2026)*

- **Lead Owner: Asad Ullah Zafar (`CS-24082`)**
- **Mapped Requirements:** `FR-5`, `FR-6`, `FR-7`, `FR-8` UI, `FR-9` UI, `Usability NFR`
- **Tasks:**
    - `1.5.1` **Login View & Role-Protected Routing (`FR-8`):** Build `/login` with JWT storage, Axios auth interceptors, and role-aware navigation for `Analyst`, `TeamLead`, and `Admin` users.
    - `1.5.2` **Prioritized Analyst Review Queue (`FR-5`):** Build `/queue` with tabs for `Awaiting Review` (`AWAITING_ANALYST` + `NEEDS_MANUAL_TRIAGE`), `Escalated` (`ESCALATED`), and `Closed` (`CLOSED`), sorted by effective severity (`CRITICAL` → `LOW`) and `ingested_at ASC`, with severity/rule/search filters connected live to PostgreSQL via FastAPI.
    - `1.5.3` **Split-Pane Alert Detail & Evidence Trace Modal (`FR-5`):** Render the structured `EvidencePack` tree with visual highlights on every dot-notation key in `cited_evidence_keys`, MITRE ATT&CK badges, raw Wazuh JSON log viewer, Safety Floor / Abstention warning banners, and injected few-shot exemplars.
    - `1.5.4` **Decision Capture & Database Audit Integration (`FR-6`):** Connect `Approve` (`≤ 2 clicks`), `Override` (`≥ 10 char` mandatory rationale with inline HTTP `422` error banner, triggering `FR-4` exemplar creation), `Escalate`, `Retry AI`, `Manual Close`, and Team Lead `Resolve Escalation` directly to atomic PostgreSQL transactions in `alerts`, `analyst_decisions_audit`, and `feedback_exemplars`.
    - `1.5.5` **Team Lead Agreement Analytics, CSV Export & Governance Views (`FR-7`, `FR-4`, `FR-9`):** Build `/analytics` charts (`% Approved`, `% Overridden`, Safety Floor/Abstention counts, daily Override Rate Trendline), one-click streaming CSV download, Exemplar Revocation table (`is_active` toggle), and Host Criticality Registry UI.

`WP-6`: Design Document, Adversarial Benchmark (`FR-10`), Testing Document, Final CD & Module Form *(Weeks 11–15 | Nov 02 – Nov 27, 2026)*

- **Owners:** All Three Members (Coordinated Submission Deliverables #4, #5, and #6)
- **Tasks:**
    - `1.6.1` **Deliverable #4 — Design Document (`Due: 02 Nov 2026`):** Compile architectural design, UML/Mermaid sequence & class diagrams, database schema DDL, API specifications, and UI wireframes.
    - `1.6.2` **Adversarial & Baseline Evaluation Runner (`FR-10`):** Execute the 30-payload prompt-injection benchmark (`0%` bypass target) and compare `BASELINE_RAW_LLM` vs. `SENTRAX_FULL` (Abdur Rehman & Atta Ur Rehman, integrated with DB/UI by Asad Ullah Zafar).
    - `1.6.3` **Deliverable #5 — Testing Document & Final Project CD (`Due: 25 Nov 2026`):** Execute unit, integration, RBAC, immutability, and `5 alerts/sec` load tests; package complete source

code, Docker Compose setup, database seeders, and Testing Document onto CD.

- `1.6.4` **Deliverable #6 — Module Distribution Form & Functionality Report (`Due: 27 Nov 2026`):** Submit signed module ownership breakdown, core features checklist, and working vs. non-working functionality declaration.

4.3 Repository & Module Implementation Blueprint

```
c:\projects\SentraX\
├── docker-compose.yml                  # [Atta & Asad] Orchestrates postgres:16,
backend, and frontend
├── .env.example                        # Environment template (WAZUH_*,
LLM_PROVIDER, JWT_SECRET, DB_URL)
├── docs/                               # Project documentation & deliverables
│   ├── 2026 Project deadlines.pdf      # Official Fall 2026 Deliverables &
Deadlines
│   ├── Sentra-X_Proposal_Updated.pdf   # Deliverable #1 (Submitted Sept 07, 2026)
│   ├── sentraX_srs.md                  # Deliverable #2 (Submitted Sept 28, 2026;
updated v1.3)
│   ├── sentraX_project_plan.md         # Deliverable #3 (Submitted Oct 12, 2026)
│   └── gemini.md                       # Persistent Engineering Context Memory
├── backend/                            # [Atta & Abdur; DB Models/Integration by
Asad]
│   ├── Dockerfile
│   ├── requirements.txt
│   ├── alembic/                        # [Asad] PostgreSQL 16 migrations + audit
immutability trigger
│   ├── app/
│   │   ├── main.py                     # [Atta] FastAPI app, CORS, lifespan Wazuh
poller
│   │   ├── database.py                 # [Asad] SQLAlchemy 2.0 session &
PostgreSQL connection pool
│   │   ├── models/                     # [Asad] 7 ORM models (users, assets,
alerts, packs, assessments, audit, exemplars)
│   │   ├── schemas/                    # [Asad] Shared Pydantic v2 DTOs matching
Frontend TypeScript types
│   │   ├── services/
│   │   │   ├── ingestion_service.py    # [Atta] FR-1: Wazuh poller/webhook, noise
filter (level >= 4), dedup
│   │   │   ├── auth_service.py         # [Atta] FR-8: Bcrypt (cost >= 10), JWT &
RBAC guards
│   │   │   ├── calibration_service.py  # [Atta] FR-4: Sub-50ms tiered exemplar
retrieval & revocation
│   │   │   ├── evidence_assembler.py   # [Abdur] FR-2: Wazuh log parser, MITRE,
24h correlation & valid_dot_keys
│   │   │   ├── prompt_sanitizer.py     # [Abdur] FR-2 / NFR-3: Delimiter
stripping & injection_suspect scanner
│   │   │   ├── llm_adapter.py          # [Abdur] FR-2 / NFR-6: Gemini & Claude
adapters + 3x retry backoff
│   │   │   ├── rules_engine.py         # [Abdur] FR-3: Deterministic Abstention &
Multi-Tier Safety Floor
│   │   │   └── evaluation_runner.py    # [Abdur] FR-10: BASELINE_RAW_LLM vs.
SENTRAX_FULL & 30-payload runner
```

```
│   │   └── routers/                    # [Asad & Atta] REST API endpoints binding
Frontend to DB & Services
│   └── tests/                          # [All] Unit, safety floor boundary, DB
immutability & injection tests
└── frontend/                           # [Asad] React 19 + TypeScript + Vite +
Tailwind CSS + shadcn/ui
    ├── Dockerfile
    ├── src/
    │   ├── api/                        # [Asad] REST API client, JWT interceptor
& DB state hooks
    │   ├── context/                    # [Asad] AuthContext & RBAC route
protection
    │   ├── views/
    │   │   ├── LoginView.tsx           # [Asad] FR-8: JWT login screen
    │   │   ├── TriageQueueView.tsx     # [Asad] FR-5: Prioritized Analyst Queue,
tabs & filters
    │   │   ├── TeamLeadAnalyticsView.tsx # [Asad] FR-7: Agreement charts,
override trendlines & CSV export
    │   │   └── GovernanceView.tsx      # [Asad] FR-4 & FR-9: Exemplar revocation
& Asset Criticality UI
    │   └── components/
    │       ├── AlertDetailModal.tsx    # [Asad] FR-5: Split-pane EvidencePack
inspector & cited-key highlights
    │       └── DecisionActionPanel.tsx # [Asad] FR-6: Approve, Override,
Escalate, Retry AI & Resolve actions
```

---

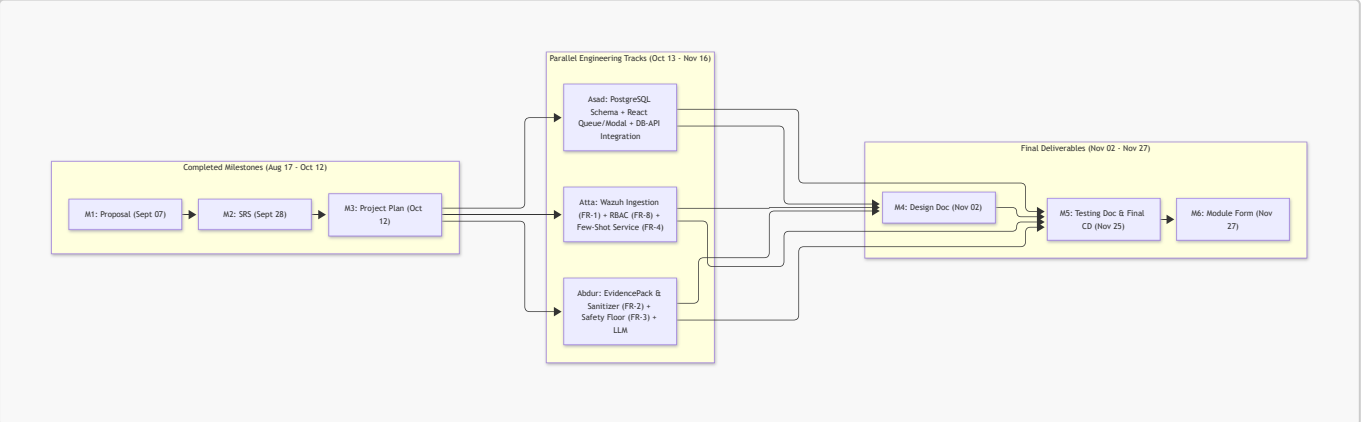
5. Project Schedule, Sprints & Milestones (15-Week Fall Semester)

5.1 Fall Semester 2026 Master Schedule (Weeks 1–15: Aug 17 – Nov 27, 2026)

Weeks & Dates	Phase / Sprint	Key Activities & Deliverables	Primary Owner(s)
Weeks 1–3 (17 Aug – 07 Sept 2026)	**Phase 1: Proposal & Novelty**	Problem formulation, literature comparison (sentra_novelty_research.pdf), 5-pillar architecture, **Deliverable #1: Project Proposal (07 Sept)**.	All (Completed)
Weeks 4–6 (08 Sept – 28 Sept 2026)	**Phase 2: Requirements (SRS)**	Elicitation of FR-1–FR-10 & NFR-1–NFR-6, System Models, 7-table ER schema, **Deliverable #2: SRS (28 Sept)** & React scaffold.	All (Completed)
Weeks 7–8 (29 Sept – 12 Oct 2026)	**Phase 3: SRS v1.3 & Project Plan**	Refinement of SRS to v1.3 (SentraX_SRS_Final.pdf), WBS & team role allocation, **Deliverable #3: Project Plan (12 Oct)**.	All (Completed)

Weeks & Dates	Phase / Sprint	Key Activities & Deliverables	Primary Owner(s)
Weeks 9–11 (13 Oct – 02 Nov 2026)	**Sprint 1: DB Layer, Wazuh Ingestion, AI Engine & Design Doc**	• **Asad:** PostgreSQL 16 schema, 7 ORM models, indexes, audit immutability trigger, API DTOs, UI wireframes & Queue/Login views. • **Atta:** Wazuh poller/webhook (FR-1), JWT RBAC (FR-8), Asset service (FR-9). • **Abdur:** EvidenceAssembler, PromptSanitizer (FR-2), RulesEngine (FR-3). • **Deliverable #4: Design Document (02 Nov)**.	Asad (DB & UI), Atta (Wazuh & Auth), Abdur (AI & Rules)
Weeks 12–13 (03 Nov – 16 Nov 2026)	**Sprint 2: Full-Stack Integration, Decision Workflows & Calibration**	• **Asad:** Full-stack integration of Prioritized Analyst Queue, Cited-Key Evidence Modal (FR-5), Decision Capture (FR-6), and Team Lead Analytics/Governance UI (FR-7) with PostgreSQL & FastAPI. • **Atta:** Tiered Few-Shot Calibration service (FR-4) & Docker Compose wiring. • **Abdur:** Gemini/Claude live integration, retry fallback (NEEDS_MANUAL_TRIAGE), and analytics SQL queries.	Asad (Frontend-DB-Backend Integration), Atta & Abdur (Backend)
Weeks 14–15 (17 Nov – 27 Nov 2026)	**Sprint 3: Evaluation (FR-10), Testing Doc, CD & Module Form**	• **All:** End-to-end testing, FR-10 30-payload adversarial benchmark & BASELINE_RAW_LLM vs. SENTRAX_FULL evaluation. • **Deliverable #5: Testing Document & Final Project in CD (25 Nov)**. • **Deliverable #6: Module Distribution Form & Core Features Details (27 Nov)**.	All Members

5.2 Critical Path & Parallel Execution Diagram



5.3 Milestone Gate Criteria (M1 to M6)

Milestone	Official Deadline	Verification Gate / Acceptance Criteria
`M1`: **Project Proposal**	**07 Sept 2026 (Mon)**	Proposal approved by Mr. Muhammad Ali Akhtar and Dr. Fakhra Aftab (`Completed`).
`M2`: **SRS Submission**	**28 Sept 2026 (Mon)**	Complete SRS specifying `FR-1`–`FR-10`, `NFR-1`–`NFR-6`, System Models, and 7-table ER schema (`Completed`).
`M3`: **Project Plan**	**12 Oct 2026 (Mon)**	SPMP v1.1 baselined with Fall 2026 deadlines and finalized 3-member module distribution (`Completed`).
`M4`: **Design Document & Core Backend/DB Gate**	**02 Nov 2026 (Mon)**	Design Document submitted; PostgreSQL 7-table schema & ORM live (Asad); Wazuh ingestion & RBAC working (Atta); Evidence Pack & Safety Floor unit-tested (Abdur).
`M5`: **Testing Document & Final Project (in CD)**	**25 Nov 2026 (Wed)**	Full React SPA integrated with Backend & PostgreSQL (Asad); live Gemini/Claude triage & `FR-10` 30-payload injection benchmark (`0%` bypass) verified; Testing Document & CD submitted.
`M6`: **Module Distribution Form & Final Audit**	**27 Nov 2026 (Fri)**	Module Distribution Form detailing each member's core features and 100% working functionality verification submitted prior to Final Exams (`30 Nov`).

6. Resource, Effort & Cost Estimation

6.1 Hardware & Software Resource Requirements

Resource Category	Specification	Owner / Usage	Cost
Development Workstations	3x Laptops (`≥ 4 vCPUs`, `8–16 GB RAM`, SSD, Docker Desktop)	Used by Asad (Frontend/DB), Atta (Wazuh/Backend), and Abdur (AI/Rules).	`$0` (Existing)
SIEM / Telemetry Source	Wazuh v4.x + Synthetic Wazuh Alert Replay Generator	Atta & Abdur — generates realistic and adversarial alerts for `FR-1` & `FR-10`.	`$0` (Open Source)
Database & Backend Stack	PostgreSQL 16, SQLAlchemy 2.0, Python 3.11+, FastAPI, Pydantic v2	Asad (Schema/ORM/Integration), Atta & Abdur (Services & Guardrails).	`$0` (Open Source)
Frontend & UI Stack	React 19, TypeScript, Vite, Tailwind CSS, shadcn/ui, Penpot	Asad — Analyst Queue, Evidence Modal, Decision Capture & Analytics UI.	`$0` (Open Source)
Hosted LLM APIs	Google Gemini API (`google-genai`) &	Abdur — Schema-constrained grounded triage inference (`FR-2`, `FR-10`).	Free Tier / Credits

Resource Category	Specification	Owner / Usage	Cost
	Anthropic Claude API (`anthropic`)		

6.2 Engineering Effort Estimation (Person-Hours Across 15 Weeks)

Across the 15-week semester (`17 Aug – 27 Nov 2026`), assuming **12 hours/week per member** (`3 members × 12 hrs/wk × 15 wks = 540 total person-hours`):

Work Package	Asad (`CS-24082`) *(Frontend & DB Integration)*	Atta (`CS-24092`) *(Wazuh & Backend Core)*	Abdur (`CS-24098`) *(Backend AI & Guardrails)*	Total Hours	% Effort
`WP-1`: Proposal, SRS & Project Plan (`S.No 1–3`)	35 hrs	35 hrs	35 hrs	**105 hrs**	19.4%
`WP-2`: PostgreSQL 16 Schema, ORM, Indexes & API Contracts	45 hrs	10 hrs	10 hrs	**65 hrs**	12.0%
`WP-3`: Wazuh Ingestion (`FR-1`), Auth/RBAC (`FR-8`), Assets (`FR-9`) & Calibration (`FR-4`)	10 hrs	80 hrs	15 hrs	**105 hrs**	19.4%
`WP-4`: Evidence Pack, Sanitizer (`FR-2`), LLM Adapters & Safety Floor (`FR-3`)	10 hrs	20 hrs	80 hrs	**110 hrs**	20.4%
`WP-5`: React SPA Queue (`FR-5`), Decisions (`FR-6`), Analytics (`FR-7`) & DB Integration	60 hrs	10 hrs	10 hrs	**80 hrs**	14.8%
`WP-6`: Design Doc (`S.No 4`), Benchmark (`FR-10`), Testing Doc + CD (`S.No 5`) & Module Form (`S.No 6`)	20 hrs	25 hrs	30 hrs	**75 hrs**	13.9%
Total Estimated Effort	**180 hrs**	**180 hrs**	**180 hrs**	**540 hrs**	**100.0%**

6.3 LLM API Quota & Operational Budget Control

1. **Mandatory Pre-Filtering (`FR-1`):** Events with `rule.level < 4` are ignored before LLM invocation, cutting unnecessary API calls by `60%–80%`.
2. **Idempotent Deduplication:** Duplicate `wazuh_alert_id` payloads are skipped at ingestion.
3. **Mock LLM Mode for Frontend & Load Testing:** Asad's frontend-to-database integration tests and Atta's `5 alerts/sec` ingestion stress tests can toggle `LLM_PROVIDER=mock` to test queue rendering

and database transactions at zero API cost, reserving live Gemini/Claude calls for end-to-end verification and Abdur's `FR-10` benchmark runs.

---

7. Risk Management Plan

7.1 Risk Assessment Methodology

Risks are rated by **Probability** (`Low`, `Medium`, `High`) and **Impact** (`Low`, `Medium`, `High`, `Critical`), with concrete technical and schedule mitigations assigned to the responsible owner.

7.2 Technical, Operational & Schedule Risk Register

Risk ID	Risk Description	Owner	Prob.	Impact	Mitigation & Contingency Strategy
`R-1`	**LLM Hallucinates Citations or Downgrades Critical Alerts:** LLM cites non-existent keys or marks a `level ≥ 10` attack as `FALSE_POSITIVE`.	Abdur (`CS-24098`)	Med	Critical	**Mitigated by `FR-3`:** `RulesEngine` verifies every key in `cited_evidence_keys` against `valid_dot_keys` and enforces the Multi-Tier Safety Floor, forcing `NEEDS_ANALYST_REVIEW`.
`R-2`	**Indirect Prompt Injection in Wazuh Logs:** Malicious strings in `full_log` or `commandLine` attempt to force `FALSE_POSITIVE`.	Abdur (`CS-24098`)	High	Critical	**Mitigated by `FR-2` & `FR-3`:** `PromptSanitizer` strips delimiters, sets `injection_suspect = true`, isolates telemetry in read-only JSON, and `FR-3` blocks benign verdicts when flagged.
`R-3`	**Frontend-Backend-Database Integration Bottleneck Late in Semester:** UI views wait on backend services to be completed.	Asad (`CS-24082`)	Med	High	**Mitigated by DB-First & Seed Data:** Asad builds the PostgreSQL schema, ORM models, and seeded test alerts first (Week 9), enabling full React queue/modal/decision integration against real DB rows while Wazuh/LLM services are wired in parallel.
`R-4`	**Full Wazuh VM Resource Contention on Local Machines:** Running full Wazuh Indexer/Dashboard consumes 8+ GB RAM.	Atta (`CS-24092`)	Med	Med	**Mitigated by Replay Service:** Atta provides a lightweight Wazuh Alert Replay utility that streams authentic Wazuh v4.x JSON alerts into `FR-1` without requiring the heavy OpenSearch indexer during daily coding.
`R-5`	**Hosted LLM API Rate Limits or Timeouts (>**	Abdur & Atta	Med	High	**Mitigated by `FR-2` & `NFR-6`:** 3x exponential backoff (`1s`, `2s`, `4s`) with

Risk ID	Risk Description	Owner	Prob.	Impact	Mitigation & Contingency Strategy
	`10s`**):** Cloud API throttling stalls live demos or testing.				deterministic fallback to `NEEDS_MANUAL_TRIAGE` (with UI `Retry AI` button) and config-only switch between Gemini and Claude.
`R-6`	**Tight Compression Between Design Doc (`Nov 02`) and Final CD (`Nov 25`):** Only 23 days separate Deliverables #4 and #5.	All Members	Med	High	**Mitigated by Parallel Sprints:** Core database, ingestion, and rules engine code is completed alongside the Design Document by `Nov 02`, leaving `Nov 03–Nov 25` focused on full-stack integration, UI polish, and testing.

8. Quality Assurance, Verification & Evaluation Plan

8.1 Multi-Layer Testing Strategy (For Nov 25 Testing Document)

1. **Database & Integration Tests (Led by Asad Ullah Zafar):**
   - Verify foreign-key integrity and cascade behavior across all 7 PostgreSQL tables (`users`, `assets`, `alerts`, `evidence_packs`, `ai_assessments`, `analyst_decisions_audit`, `feedback_exemplars`).
   - Verify PostgreSQL `BEFORE UPDATE OR DELETE` trigger blocks any mutation or deletion on `analyst_decisions_audit` (`FR-6`, `NFR-4`).
   - Verify end-to-end UI-to-Database workflows: Analyst Queue filtering/sorting (`FR-5`), Evidence Trace Modal cited-key highlighting (`FR-5`), `≤ 2 clicks` Approval, Override with `< 10 char` HTTP `422` rejection and `≥ 10 char` exemplar creation (`FR-6`, `FR-4`), Escalation Resolution, and Analytics/CSV export (`FR-7`).
2. **Wazuh Ingestion, Auth & Calibration Tests (Led by Atta Ur Rehman):**
   - Verify `rule.level < 4` noise filtering and idempotent `wazuh_alert_id` deduplication (`FR-1`).
   - Verify JWT authentication (`bcrypt cost ≥ 10`) and HTTP `403 Forbidden` enforcement across `Analyst`, `TeamLead`, and `Admin` endpoints (`FR-8`).
   - Verify `< 50ms` two-tier exemplar lookup (`Tier 1: rule_id + process_signature`, `Tier 2: rule_id + asset_criticality`) and Team Lead exemplar revocation (`FR-4`, `NFR-4`).
3. **AI Triage, Safety Floor & Adversarial Tests (Led by Abdur Rehman):**
   - Boundary unit tests for `FR-3` Safety Floor across Wazuh levels `3, 4, 7, 8, 9, 10, 11, 12, 15` and Abstention conditions (`confidence < 0.60`, empty/invalid `cited_evidence_keys`, contradictory outputs).
   - Execution of the **30-Payload Adversarial Prompt-Injection Suite** (`Appendix C` of SRS) verifying `0%` bypass rate (`NFR-3`).

8.2 Non-Functional Target Verification Matrix

NFR ID	Metric	Target Threshold	Responsible Tester	Verification Method
NFR-1	End-to-end triage latency (PENDING_REVIEW → AWAITING_ANALYST)	≤ 8.0 seconds	Abdur Rehman	Latency logging in ai_assessments across live Gemini/Claude runs.
NFR-2	Service uptime & graceful LLM timeout fallback	≥ 99.0%; 3x retry → NEEDS_MANUAL_TRIAGE	Atta Ur Rehman	Simulated LLM timeout/HTTP 503 test & Docker restart check.
NFR-3	Adversarial prompt-injection bypass rate; password security	0% bypass (0/30); bcrypt cost ≥ 10	Abdur & Atta	FR-10 30-payload adversarial test suite & hash inspection.
NFR-4	Few-shot lookup latency; CSV export speed; immutable audit	< 50ms lookup; ≤ 10s CSV; append-only DB	Asad & Atta	EXPLAIN ANALYZE on feedback_exemplars, CSV timer & trigger test.
NFR-5	Sustained ingestion throughput; 50k analytics query speed	≥ 5 alerts/sec; ≤ 5.0s analytics render	Atta & Asad	300 alerts/min replay stress test & 50k seeded rows analytics test.
NFR-6	LLM provider swap (Gemini ↔ Claude) & container startup	Config-only switch; single docker compose up	Abdur & Atta	Clean boot verification toggling LLM_PROVIDER env variable.

8.3 Research & Adversarial Evaluation Protocol (FR-10)

For the **Testing Document (Due: 25 Nov 2026)**, the EvaluationRunner (FR-10) compares:

- **BASELINE_RAW_LLM:** Raw Wazuh alert JSON sent directly to the LLM without EvidencePack enrichment, PromptSanitizer, FR-3 Safety Floor/Abstention, or FR-4 exemplars.
- **SENTRAX_FULL:** Full Sentra-X pipeline (FR-1 through FR-4).
- **Measured Metrics:** TP/FP Precision, Recall, F1, False-Negative Rate on High/Critical alerts, Abstention Accuracy on incomplete logs, Repeat Override Rate reduction across sequential similar alerts, Mean Latency (ms), and Prompt-Injection Bypass Rate (0% target).

---

9. Configuration, Change & Delivery Management

9.1 Version Control & Branching Strategy

- **main branch:** Protected stable branch containing verified database migrations, backend services, and React views.
- **Subsystem branches:** Scoped by owner and feature (asad/db-schema-orm, asad/react-queue-modal, atta/wazuh-ingestion-rbac, abdur/evidence-pack-safety-floor).

9.2 Documentation & Persistent Context Protocol

- **[sentraX_srs.md](file:///c:/projects/SentraX/docs/sentraX_srs.md):** Canonical requirements baseline.
- **[sentraX_project_plan.md](file:///c:/projects/SentraX/docs/sentraX_project_plan.md):** Canonical schedule and module distribution plan.
- **[gemini.md](file:///c:/projects/SentraX/docs/gemini.md):** Updated after every database, backend, or frontend milestone.

9.3 Final CD Submission & Module Distribution Handover Checklist (Nov 25–27)

1. **Final Project CD (`25 Nov 2026`):** Contains complete source code (`backend/`, `frontend/`, `docker-compose.yml`), database migration & seed scripts, benchmark datasets, and all formal PDF documents (Proposal, SRS, Project Plan, Design Document, Testing Document).
2. **Module Distribution Form (`27 Nov 2026`):** Signed declaration mapping **Asad Ullah Zafar (`CS-24082`)** to Frontend SPA & Database Integration, **Atta Ur Rehman (`CS-24092`)** to Wazuh Ingestion, RBAC & Feedback Calibration Backend, and **Abdur Rehman (`CS-24098`)** to Evidence Pack, Prompt Sanitizer, LLM Adapters, Safety Floor & Adversarial Benchmarking, with 100% working status across `FR-1`–`FR-10`.

---

10. Appendix: Document Revision History

Version	Date	Authors	Description of Changes
1.0	Oct 12, 2026	Asad Ullah Zafar, Atta Ur Rehman, Abdur Rehman	Initial Software Project Management & Implementation Plan (SPMP) draft.
1.1	Oct 12, 2026	Asad Ullah Zafar (CS-24082), Atta Ur Rehman (CS-24092), Abdur Rehman (CS-24098)	Aligned schedule and milestones with official **Fall Semester 2026 Project Deliverables & Deadlines (`17 Aug – 27 Nov 2026`)** and updated work distribution: **Asad Ullah Zafar** assigned Frontend functionality (Analyst Queue, Evidence Modal, Decision Capture, Analytics UI) and PostgreSQL Database integration with Frontend/Backend; **Atta Ur Rehman** and **Abdur Rehman** assigned Wazuh SIEM integration and Backend services (`FR-1`–`FR-4`, `FR-8`–`FR-10`).